JSS MAHAVIDYAPEETHA
JSS SCIENCE AND TECHNOLOGY UNIVERSITY
SRI JAYACHAMARAJENDRA COLLEGE OF ENGINEERING

- Constituent College of JSS Science and Technology University
- Approved by A.I.C.T.E
- Governed by the Grant-in-Aid Rules of Government of Karnataka
- Identified as lead institution for World Bank Assistance under TEQIP Scheme

Question Bank 22EE110
Unit-III
2022-2023

Branch : EEE
Subject: Basics of Electrical Engineering

Chapter 3: DC Circuits and Electromagnetism

1. Define the following terminologies in D.C Circuits with their unit.
- **Current**
- **Charge**
- **Electric Potential**
- **Potential Difference**
- **Resistance**
- **Electromotive Force**
- **Electric Power**
- **Electrical Energy**

2. What are the two classification of elements in D.C circuits. Explain with examples.
3. State Ohm's law. List the limitations of Ohm's law.
4. Briefly explain the series and parallel circuits with necessary circuit diagram and equations.
5. Illustrate the Kirchoff's laws.
6. Problems on Series and Parallel circuits.
7. Problems on Mesh Analysis and Node Analysis.
8. Define the following:
- **Magnetism**
- **Magnetic field**
- **Magnetic flux**
- **Magnetic flux density**
- **Magnetic permeability**
- **Intensity of Magnetization**
- **Magnetic field strength/Field Intensity**

- **Susceptibility**
- **MMF**
- **Reluctance**

9. Write the difference between magnetic and electric circuits.

10. Explain the Faraday's Laws of Electromagnetic Induction.

11. Explain the Lenz's Law and Flemming's law of Electromagnetic Induction.

12. With necessary equations, briefly explain the following:

- **Self inductance**
- **Mutual inductance.**
- **Coefficient of Coupling**

13. Problems on:

- **Self inductance**
- **Mutual inductance.**
- **Coefficient of Coupling**